

## Tutorial Week 1 : Foundations I

Answer the following questions in your own words.

### SAQ

1. For each of the following, Illustrate the differences and provide sample code snippets to illustrate the differences:
  - a. A procedural language and an object-oriented language.
  - b. A procedural language and a generic programming language.
2. The codes below are in procedural C++.
  - a. Modify it to reflect procedural C.
  - b. Modify the original codes in other versions of C++. Suggest and use other methods of passing an array to the function.
  - c. Modify the original codes to reflect generic programming.

```
#include<iostream>
using namespace std;

const int SIZE = 5;

void display(int arr[])
{
    cout << "Values in the array " << endl;
    for(int i = 0; i < SIZE; i++)
    {
        cout << arr[i] << "\t";
    }
}

void readInput(int a[])
{
    //use the normal for loop to prompt user and store values in a
    for(int i = 0; i < SIZE; i++)
    {
        cout << "Enter value for element " << i + 1 << ": ";
        cin >> a[i];
    }
}

int main()
{
    int arr[SIZE] = {0};
    readInput(arr);
    display(arr);
}
```